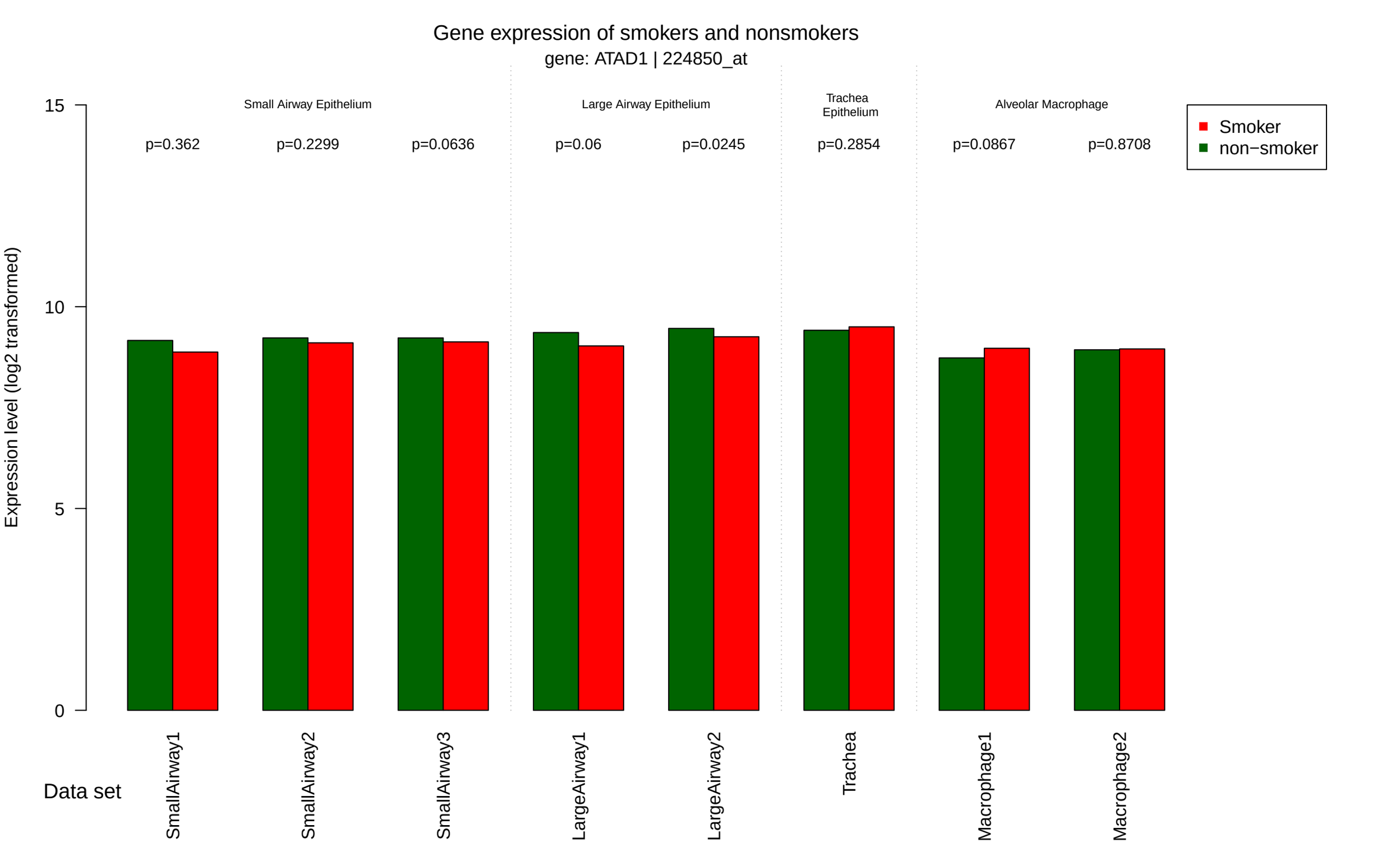
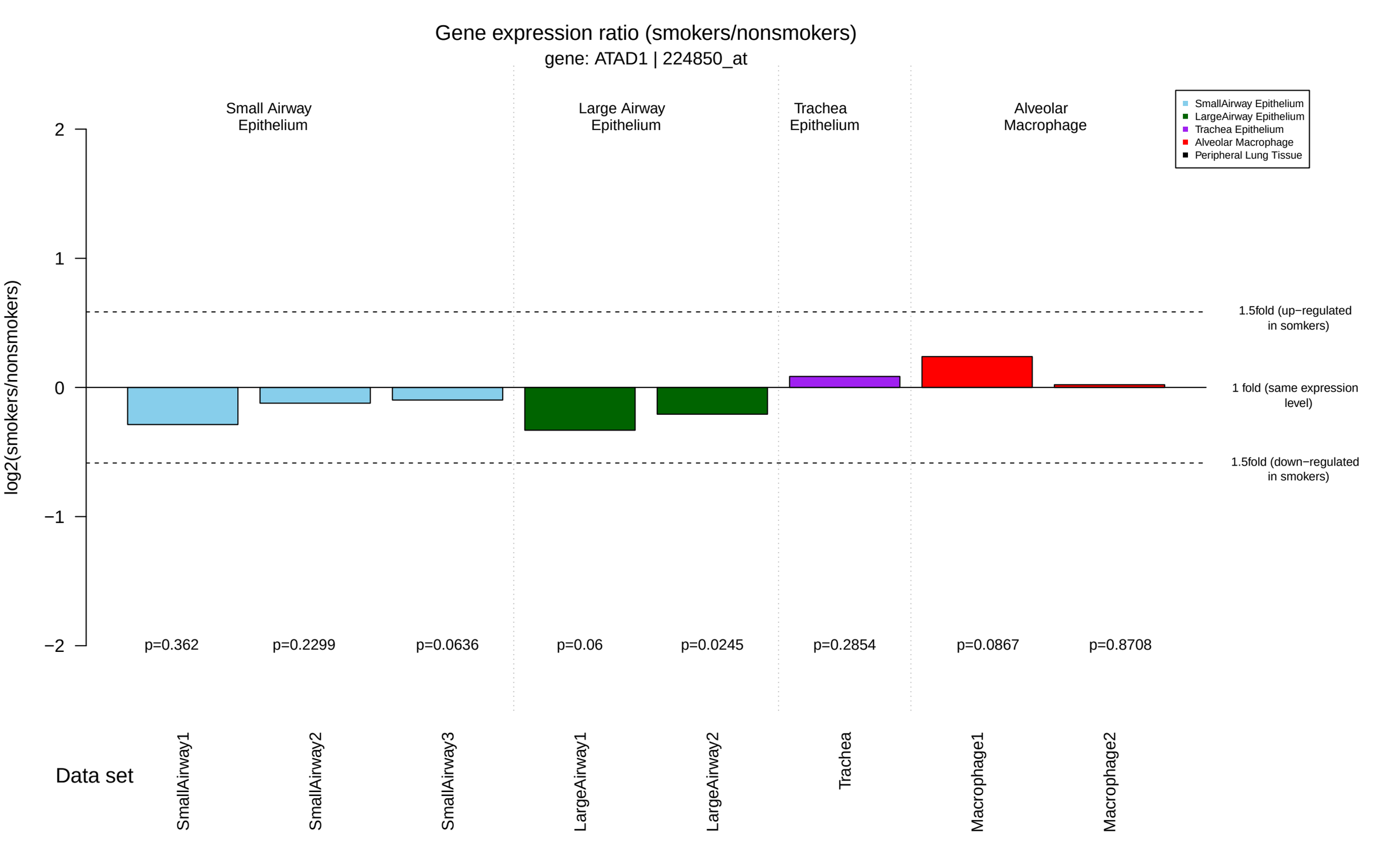
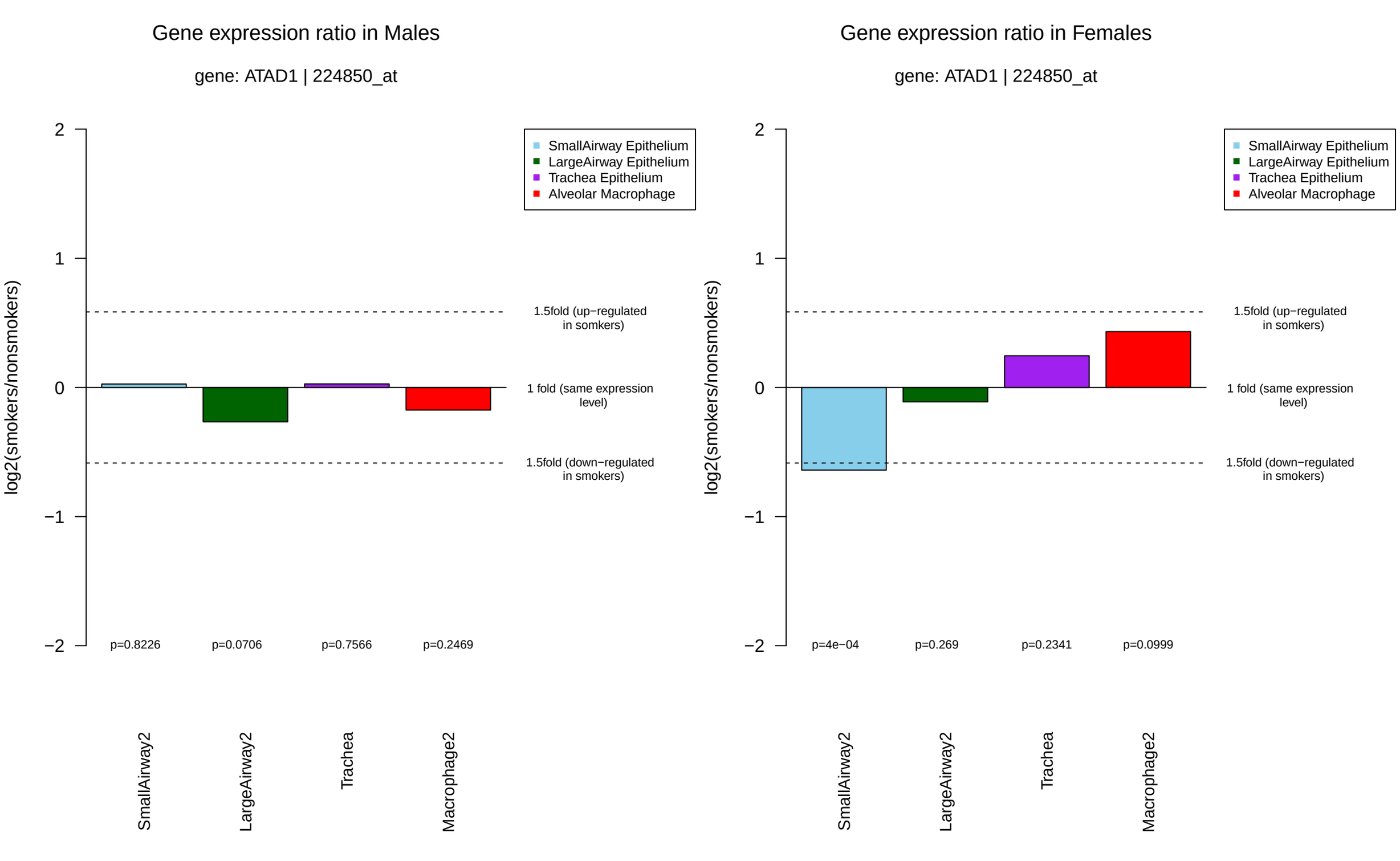
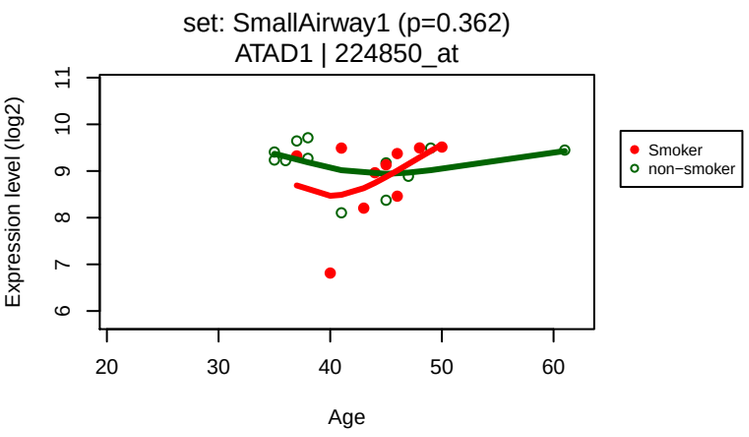
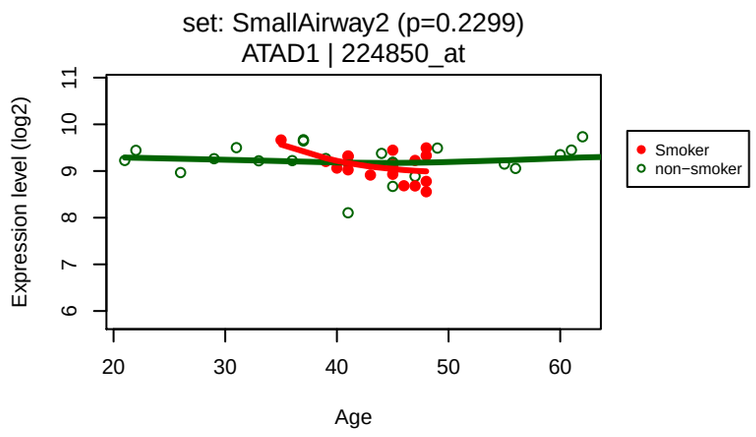
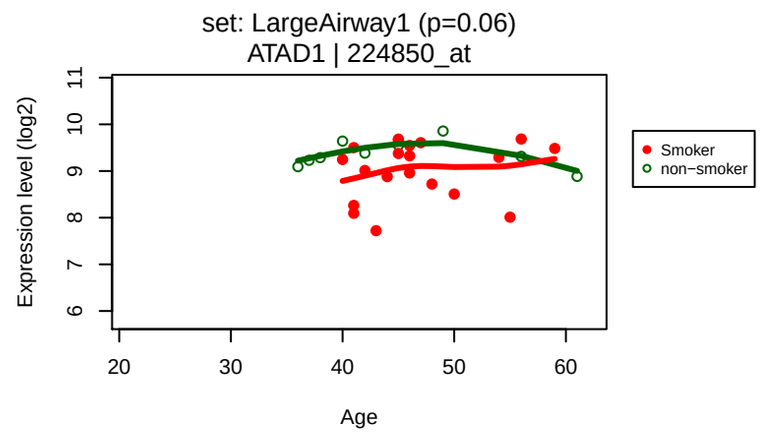

Gene expression of smokers and nonsmokers
gene: ATAD1 | 224850_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.362
p=0.2299
p=0.0636
p=0.06
p=0.0245
p=0.2854
p=0.0867
p=0.8708
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

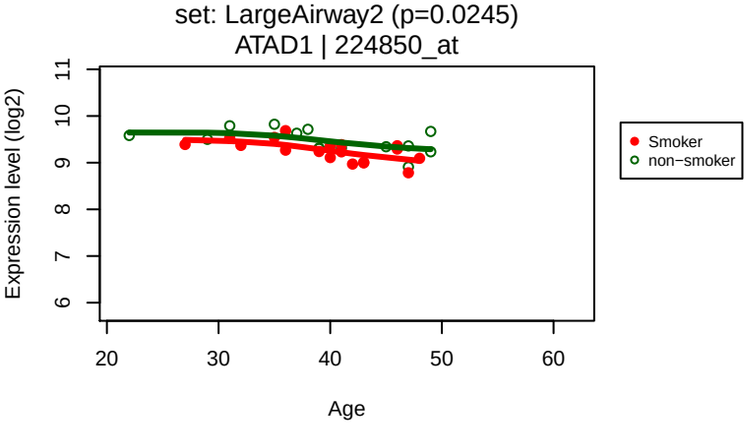

Gene expression ratio (smokers/nonsmokers)
gene: ATAD1 | 224850_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.362
p=0.2299
p=0.0636
p=0.06
p=0.0245
p=0.2854
p=0.0867
p=0.8708
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

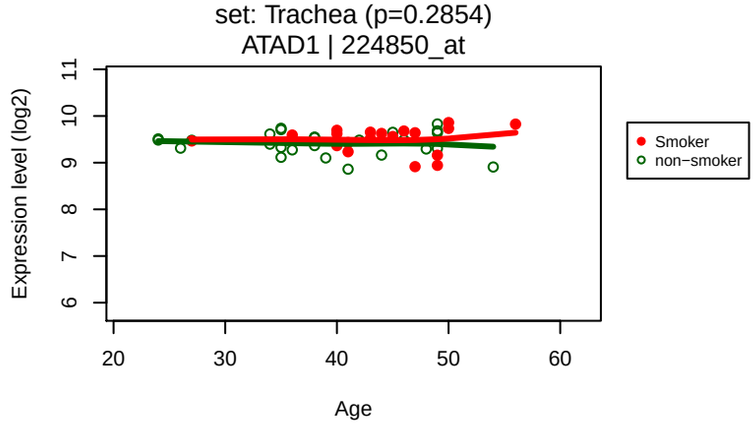

Gene expression ratio in Males
gene: ATAD1 | 224850_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8226
p=0.0706
p=0.7566
p=0.2469
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATAD1 | 224850_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.269
p=0.2341
p=0.0999
SmallAirway2
LargeAirway2
Trachea
Macrophage2

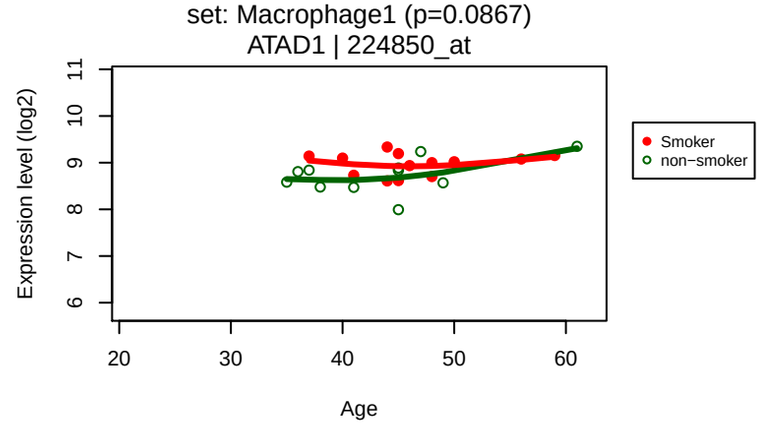

set: SmallAirway1 (p=0.362)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker

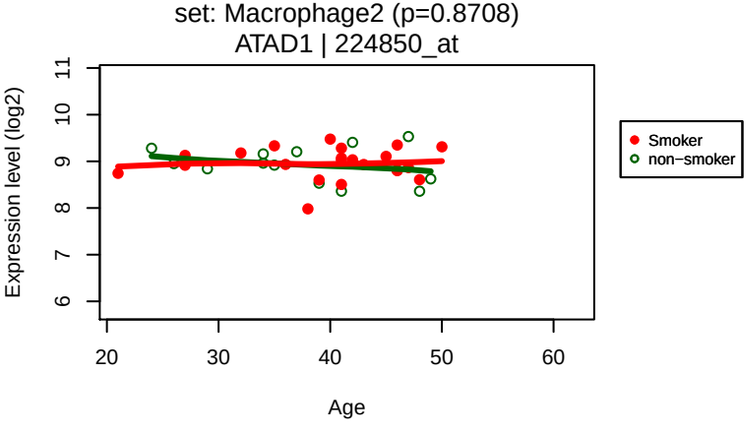

set: SmallAirway2 (p=0.2299)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker

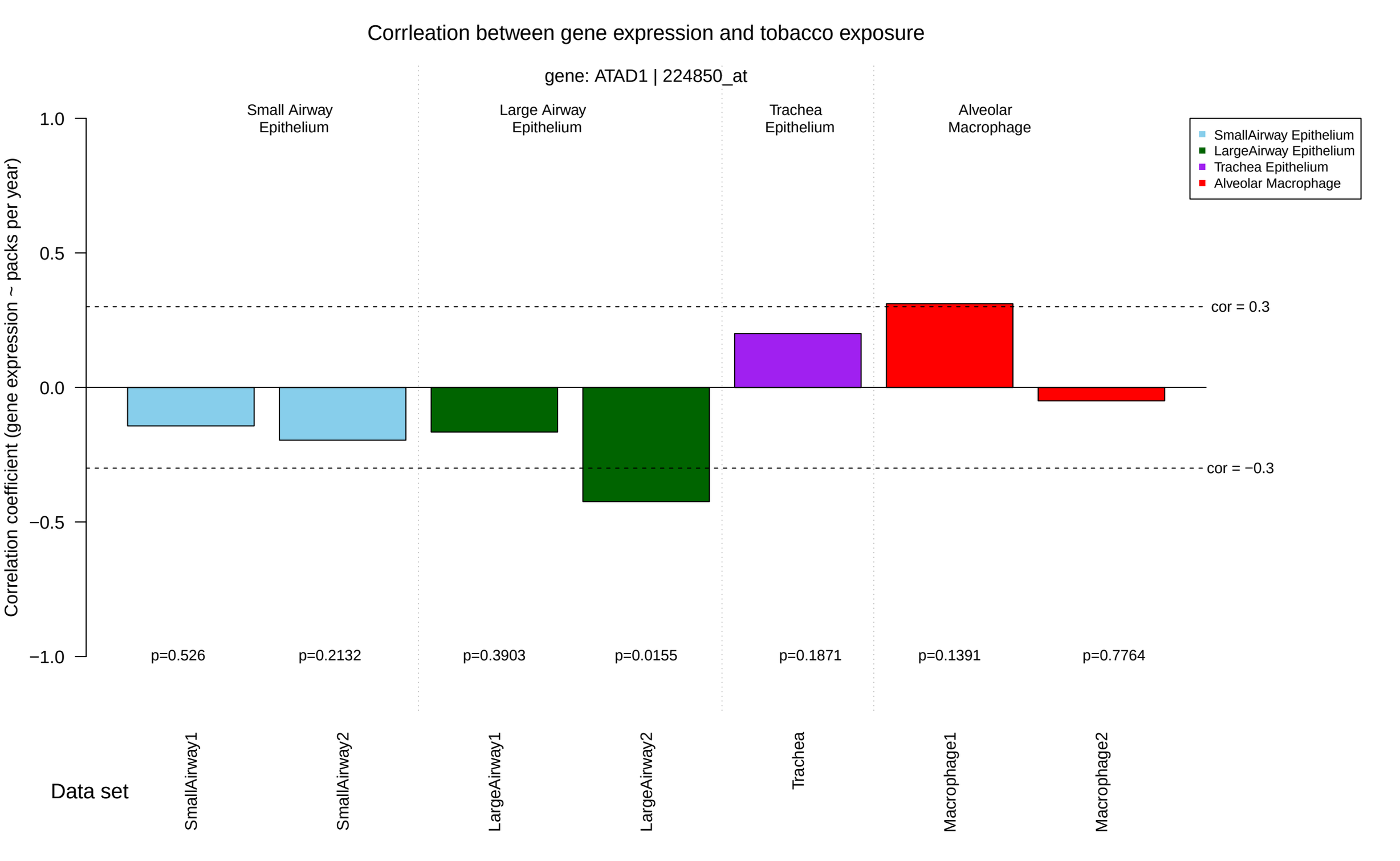

set: LargeAirway1 (p=0.06)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0245)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2854)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0867)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8708)
ATAD1 | 224850_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATAD1 | 224850_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.526
p=0.2132
p=0.3903
p=0.0155
p=0.1871
p=0.1391
p=0.7764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2